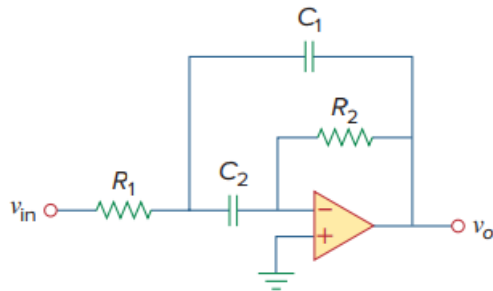


Problem

In the op amp circuit of Fig. 8.112, determine $v_o(t)$ for $t > 0$. Let $v_{in} = u(t)$ V, $R_1 = R_2 = 10$ k Ω , $C_1 = C_2 = 100$ μ F.

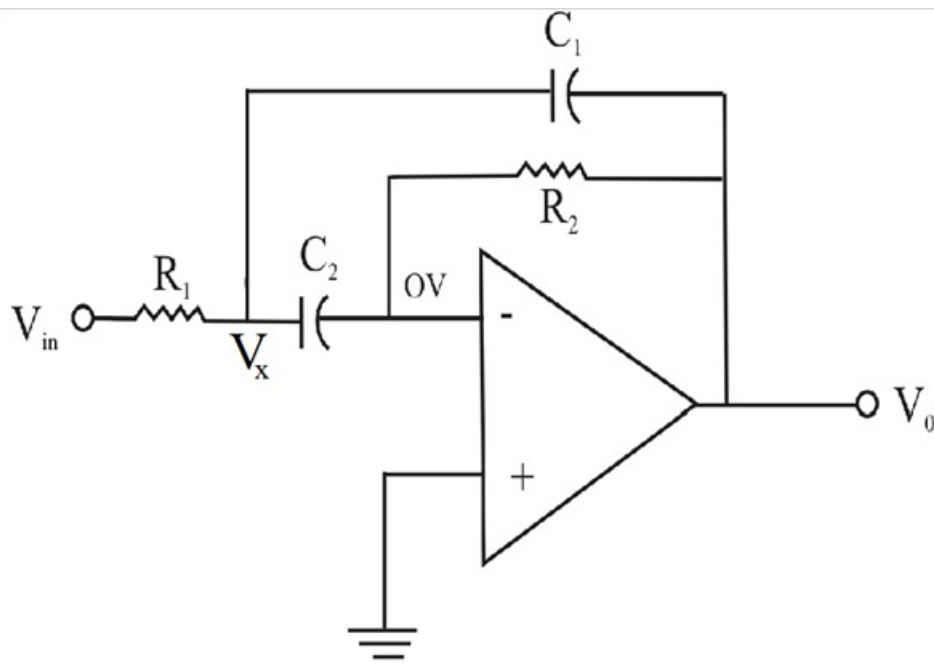
Figure 8.112:



Step-by-step solution

Step 1 of 7

Given circuit diagram:



Step 2 of 7

By applying KCL at node V_x we can write

$$\frac{V_x - V_{in}}{R_1} + C_1 \frac{d(V_x - V_o)}{dt} + C_2 \frac{d(V_x - 0)}{dt} = 0$$

$$V_x - V_{in} + R_1 C_1 \frac{d(V_x - V_o)}{dt} + R_1 C_2 \frac{d(V_x - 0)}{dt} = 0$$

$$V_x + R_1 C_1 \frac{dV_x}{dt} + R_1 C_2 \frac{dV_x}{dt} - R_1 C_1 \frac{dV_o}{dt} = V_{in}$$

$$V_x + \frac{dV_x}{dt} (R_1 C_1 + R_1 C_2) - R_1 C_1 \frac{dV_o}{dt} = V_{in} \dots\dots\dots (1)$$

Step 3 of 7

By applying KCL at inverting terminal we can write

$$C_2 \frac{d(0 - V_x)}{dt} + \frac{0 - V_o}{R_2} = 0$$

$$-R_2 C_2 \frac{dV_x}{dt} - V_o = 0$$

$$R_2 C_2 \frac{dV_x}{dt} = -V_o$$

$$\frac{dV_x}{dt} = -\frac{1}{R_2 C_2} V_o \dots\dots\dots (2)$$

Step 4 of 7

By integrating both sides we can get

$$R_2 C_2 V_x = -\int V_o dt$$

$$V_x = -\frac{1}{R_2 C_2} \int V_o dt \dots\dots\dots (3)$$

From equations (2) and (3) we can write equation (1) as below

$$-\frac{1}{R_2 C_2} \int V_o dt + \left(-\frac{1}{R_2 C_2} V_o \right) (R_1 C_1 + R_1 C_2) - R_1 C_1 \frac{dV_o}{dt} = V_{in}$$

By differentiating both sides with respect to t we can get

$$-\frac{V_o}{R_2 C_2} + \left(-\frac{1}{R_2 C_2} \frac{dV_o}{dt} \right) (R_1 C_1 + R_1 C_2) - R_1 C_1 \frac{d^2 V_o}{dt^2} = \frac{dV_{in}}{dt}$$

$$R_1 C_1 = R_1 C_2 = R_2 C_2 = 10k \times 100\mu = 1$$

Now the above equation becomes

$$-V_o - \left(\frac{dV_o}{dt} \right) (2) - \frac{d^2 V_o}{dt^2} = \frac{dV_{in}}{dt}$$

$$\frac{d^2 V_o}{dt^2} + 2 \frac{dV_o}{dt} + V_o = -\frac{dV_{in}}{dt}$$

Step 5 of 7

From this we can write the characteristic equation as $s^2 + 2s + 1 = 0$

$$(s + 1)^2 = 0$$

$$s = -1, -1$$

Now we can write expression for output

$$v_o(t) = (Ae^{-t} + Bte^{-t}) + V_f$$

Where V_f is the output final voltage

At $t = \infty$ both the capacitors act as open circuit, so

$$V_f = V(\infty) = 0V$$

$$v_o(t) = (Ae^{-t} + Bte^{-t})$$

$$v_o(t) = (A + Bt)e^{-t} \dots\dots (4)$$

Step 6 of 7

Input is step function $v_{in} = u(t)$ V and all capacitor initial voltages are equal to 0V.

Therefore $v_o(0) = 0V$

From equation (4) we can write

$$v_o(0) = A$$

$$A = 0$$

$$v_o(t) = Bte^{-t} \text{ V} \dots\dots\dots (5)$$

$$\frac{dv_o(t)}{dt} = B(e^{-t} - te^{-t}) \dots\dots\dots (6)$$

Step 7 of 7

We know that

$$\frac{V_x - V_{in}}{R_1} + C_1 \frac{d(V_x - V_o)}{dt} + C_2 \frac{d(V_x - 0)}{dt} = 0 \dots\dots\dots (7)$$

At $t = 0^+$, $V_x(0^+) = 0V$ and $V_{in}(0^+) = 1V$

Now equation (7) becomes

$$\frac{0 - 1}{R_1} + C_1 \frac{d(0 - V_o(0^+))}{dt} + C_2 \frac{d(0 - 0)}{dt} = 0$$

$$\frac{dV_o(0^+)}{dt} = -\frac{1}{R_1 C_1}$$

$$\frac{dV_o(0^+)}{dt} = -1$$

Now equation (6) becomes

$$-1 = B(e^{-0} - (0)e^{-0})$$

$$B = -1$$

Therefore from equation (5) we can write the final output expression as

$$\boxed{v_o(t) = -te^{-t}u(t) \text{ V}}$$